Q1. A textile fibre must have a length at least how many times its width?

A) 10 B) 50 C) 100 D) 500

Q2. The ideal length-to-breadth ratio of a textile fibre is greater than:

A) 50:1 B) 100:1 C) 1000:1 D) 10:1

Q3. The essential tensile strength range of a textile fibre is:

A) 0.5–1.0 g/d B) 1.5–5.0 g/d C) 6–10 g/d D) 10–15 g/d

Q4. Which is a secondary (desirable) property?

A) Uniformity B) Cohesiveness C) Moisture regain D) Staple length

Q5. Continuous conversion of a fibre-forming material into filaments without breaking is called:

A) Cohesiveness B) Spinnability C) Resilience D) Lustre

Q6. Cohesiveness of fibres is NOT due to:

A) Surface friction B) Crimp C) Natural twists D) Lustre

Q7. Kapok is a:

A) Bast fibre B) Seed hair fibre C) Leaf fibre D) Mineral fibre

Q8. Sisal and abaca are:

A) Leaf fibres B) Bast fibres C) Seed hairs D) Husk fibres

Q9. Coir is obtained from the:

A) Stem B) Leaf C) Seed hair D) Husk/fruit

Q10. Asbestos is classified under:

A) Animal fibres B) Regenerated fibres C) Mineral fibres D) Synthetic polymers

Q11. Cuprammonium rayon is a:

A) Regenerated protein fibre B) Regenerated cellulosic fibre C) Polyamide D) Mineral fibre

Q12. Polyurethane (spandex) belongs to which sub-class?

A) Elastomeric B) Polyolefin C) Polyamide D) Acrylic

Q13. A flat, twisted ribbon-like fibre with helical convolutions and a central lumen is:

A) Jute B) Linen C) Wool D) Cotton

Q14. A straight cylindrical fibre with transverse cross-markings and a narrow lumen is:

A) Linen B) Cotton C) Silk D) Viscose

Q15. Overlapping surface scales (cuticle) are seen in:

A) Silk B) Wool C) Nylon D) Polyester

Q16. A triangular (trilobal) cross-section with rounded corners is typical of:

A) Wool B) Cotton C) Silk D) Acrylic

Q17. A serrated cross-section with fine longitudinal striations indicates:

A) Polyester B) Nylon C) Silk **D) Viscose rayon**

Q18. A dog-bone or kidney-bean cross-section is characteristic of:

A) Polyester B) Acrylic C) Nylon D) Cotton

Q19. Cellulosic fibres on burning smell like:

A) Burning paper B) Burning hair C) Celery D) Sweet aromatic odour

Q20. The residue of burnt wool or silk is:

A) Light feathery ash B) Hard cream bead C) Dark, crushable bead D) Hard black crust

Q21. A fibre that melts, gives a celery-like odour and leaves a hard cream bead is:

A) Polyester B) Nylon C) Acrylic D) Wool

Q22. A smoky yellow flame with a sweet aromatic odour indicates:

A) Nylon B) Acrylic C) Cotton D) Polyester

Q23. Which fibre has the lowest density?

A) Polypropylene B) Nylon 6,6 C) Acrylic D) Wool

Q24. In a density gradient column, a fibre:

A) Dissolves B) Burns C) Stops where the liquid density equals its own D) Melts

Q25. 75% cold H_2SO_4 dissolves completely:

A) Wool B) Cotton/Linen C) Polyester D) Nylon

Q26. 5% hot NaOH dissolves completely:

A) Cotton B) Polyester C) Acrylic D) Wool

Q27. The specific solvent for Nylon 6,6 is:

A) 85% formic acid B) Dimethylformamide C) Acetone D) 60% cold H_2SO_4

Q28. Acetate dissolves rapidly in:

A) Formic acid B) Nitrobenzene C) 100% acetone D) DMF

Q29. The moisture regain of cotton is:

A) 4.5% B) 8.5% C) 11% D) 16%

Q30. The wet strength of cotton:

A) Decreases by 50% B) Is unchanged C) Decreases by 15–20% D) Increases by 10–20%

Q31. Mercerization uses NaOH of strength:

A) 1–5% B) 5–10% C) 18–25% D) 40–50%

Q32. Mercerization changes the crystal structure from:

A) Cellulose I to II B) Cellulose II to I C) Cellulose I to III D) Crystalline to amorphous

Q33. Acid hydrolysis of cotton produces:

A) Oxycellulose B) Hydrocellulose C) Keratin D) Fibroin

Q34. The lignin content of jute is approximately:

A) 1–2% B) 3–5% C) 25–30% D) 10–15%

Q35. The elongation at break of jute is:

A) 1.0–2.0% B) 6–9% C) 25–40% D) 15–30%

Q36. Jute yellows in UV light due to oxidation of:

A) Cellulose B) Hemicellulose C) Lignin D) Pectin

Q37. The dry tenacity of linen is:

A) 1.5–2.5 g/d B) 5.5–6.5 g/d C) 2.2–3.5 g/d D) 1.0–1.7 g/d

Q38. Linen is more resistant than cotton to:

A) Hot concentrated acids B) Warm dilute acids C) Chlorine bleach D) Cold dilute acids

Q39. The moisture regain of wool is:

A) 8.5% B) 11% C) 16–18% D) 0.4%

Q40. Elastic recovery of more than 99% at low extension is shown by:

A) Wool B) Cotton C) Jute D) Linen

Q41. Wool is composed of:

A) Fibroin B) Keratin with disulfide cross-links C) Cellulose D) Lignin

Q42. The safe bleaching agent for wool is:

A) Chlorine B) Hypochlorite C) Concentrated acid D) Hydrogen peroxide

Q43. In silk, the two filaments are held together by:

A) Sericin B) Keratin C) Lignin D) Pectin

Q44. Silk fibroin is rich in:

A) Cystine B) Cellulose C) Alanine, glycine and serine D) Terephthalate

Q45. The length of a silk filament is about:

A) 1–4 m B) 300–1200 m C) 3–40 cm D) 20–80 cm

Q46. Viscose rayon loses about what share of its strength when wet?

A) 5% B) 10% C) 20% D) 50%

Q47. Viscose rayon has which polymorphic structure?

A) Cellulose I B) Keratin C) Cellulose II D) Fibroin

Q48. Viscose has high affinity for:

A) Direct, reactive and vat dyes B) Disperse dyes C) Basic dyes D) Acid dyes only

Q49. The dry tenacity of HT rayon is:

A) 2.0–2.6 g/d B) 3.5–6.0 g/d C) 1.0–1.7 g/d D) 4.5–7.5 g/d

Q50. A major use of HT rayon is:

A) T-shirts B) Bedsheets C) Surgical cotton D) Tyre cord

Q51. HT rayon is produced by:

A) Lowering the degree of polymerization B) Spinning without stretch C) Modifying the spinning bath and applying high stretch D) Melt spinning

Q52. Polyester (PET) is made from:

A) Ethylene glycol and terephthalic acid B) Hexamethylene diamine and adipic acid C) Acrylonitrile D) Vinyl acetate

Q53. The melting point of polyester is about:

A) 70–80 °C B) 255–265 °C C) 120–130 °C D) 180 °C

Q54. Polyester is dyed with:

A) Basic dyes B) Direct dyes C) Vat dyes D) Disperse dyes at about 130 °C

Q55. The moisture regain of polyester is:

A) 4.0% B) 1.5% C) 0.4% D) 8.5%

Q56. Nylon 6,6 is produced from:

A) Ethylene glycol and terephthalic acid B) Hexamethylene diamine and adipic acid C) Acrylonitrile D) Vinyl acetate

Q57. Nylon 6,6 shows what elastic recovery at low extension?

A) 100% B) 74% C) 50% D) 25%

Q58. The moisture regain of nylon 6,6 is:

A) 0.4% B) 1.5–2.0% C) 8.5% D) 4.0–4.5%

Q59. Acrylic fibre is:

A) A condensation polymer with ≥ 50% acrylonitrile B) A natural polymer C) An addition polymer with ≥ 85% acrylonitrile D) A regenerated cellulose

Q60. Acrylic fibres are generally dyed with:

A) Basic (cationic) dyes B) Disperse dyes C) Vat dyes D) Direct dyes